

SESSION 5

AI Architecture on Azure — Foundry Portal Hands-On Lab

Scenario: The BFL Loan Product Assistant — Grounding Fundamentals

Participant Lab Guide

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Lab Overview & Learning Outcomes

By the end of this lab, you will be able to work confidently in the Microsoft Foundry portal for enterprise-grade activities. Specifically, you will be able to:

- Provision a Foundry project, resource group, and model deployment in a specific region
- Navigate the Foundry portal with confidence — Model catalog, Deployments, Playground, Management Center, and access control
- Apply a structured decision framework to select the right model for a given business scenario
- Use the Chat Playground to test a scenario-based assistant
- Write and refine system instructions to control assistant behavior and scope
- Manually ground a model's answers using reference documents, and recognize why this approach doesn't scale — setting up Session 6's automated Retrieval-Augmented Generation (RAG) content

Scenario Briefing: The BFL Loan Product Assistant

BFL is piloting an internal assistant that helps relationship managers answer questions about BFL's loan products. Today, you will build a first version of this assistant, discover why an ungrounded model gives unreliable answers, and manually ground it using reference policy documents. In Session 6, you will replace this manual process with Azure AI Search and automated RAG.

Before You Begin

- You already hold Contributor at the subscription level on the shared training subscription — no additional role request is needed to complete this lab.
- Locate your participant number and assigned region in the table below before starting Exercise 1.
- Two synthetic reference documents have been provided alongside this guide: BFL_SecureHome_Loan_Policy.txt and BFL_FlexiBiz_Loan_Policy.txt. These are fictional products created for training purposes only.

Naming convention reminder

All resources you create in this lab use the trn <participant-number> prefix, consistent with every lab so far in this program.

Participant Assignment Table

Participant #	Region	Resource Group	Project Name	Deployment Name
01	East US	trn_01-rg	trn_01-foundry	trn_01-gpt4o
02	East US	trn_02-rg	trn_02-foundry	trn_02-gpt4o
04	East US	trn_04-rg	trn_04-foundry	trn_04-gpt4o
05	East US	trn_05-rg	trn_05-foundry	trn_05-gpt4o
06	East US 2	trn_06-rg	trn_06-foundry	trn_06-gpt4o
07	East US 2	trn_07-rg	trn_07-foundry	trn_07-gpt4o
08	East US 2	trn_08-rg	trn_08-foundry	trn_08-gpt4o
09	East US 2	trn_09-rg	trn_09-foundry	trn_09-gpt4o
10	East US 2	trn_10-rg	trn_10-foundry	trn_10-gpt4o
11	Central US	trn_11-rg	trn_11-foundry	trn_11-gpt4o
12	Central US	trn_12-rg	trn_12-foundry	trn_12-gpt4o

13	Central US	trn_13-rg	trn_13-foundry	trn_13-gpt4o
14	Central US	trn_14-rg	trn_14-foundry	trn_14-gpt4o
15	Central US	trn_15-rg	trn_15-foundry	trn_15-gpt4o
16	North Central US	trn_16-rg	trn_16-foundry	trn_16-gpt4o
17	North Central US	trn_17-rg	trn_17-foundry	trn_17-gpt4o
18	North Central US	trn_18-rg	trn_18-foundry	trn_18-gpt4o
19	North Central US	trn_19-rg	trn_19-foundry	trn_19-gpt4o
20	North Central US	trn_20-rg	trn_20-foundry	trn_20-gpt4o

Exercise 1 — Provision Your Foundry Project

Goal: create your own isolated project, in your assigned region, and deploy GPT-4o into it.

- Sign in to the Foundry portal at ai.azure.com with your training-subscription identity.
- Select Create new project. In Advanced options, set the Resource group to your assigned trn_<##>-rg (create it if it doesn't already exist) and the Location to your assigned region from the table above.
- Name the project exactly as shown in your assignment row (trn_<##>-foundry), then select Create, and wait for provisioning to complete.
- Once inside your project, open Model catalog, search for "GPT-4o", and select Deploy.
- Choose the Global Standard deployment type. Name the deployment exactly as shown in your assignment row (trn_<##>-gpt4o), accept the default rate limit and content filter, and select Deploy.
- Confirm the deployment status shows Succeeded before continuing to Exercise 2.

If something fails

If deployment fails with a quota error, flag your trainer immediately rather than retrying repeatedly — this may indicate your region's shared quota pool needs attention.

Exercise 2 — Foundry Portal Navigation Scavenger Hunt

Goal: build comfort navigating the portal areas you'll return to throughout this program. Work through the checklist below and note your findings — your trainer will spot-check a few answers.

#	Portal Area	What to find / note
1	Project Overview page	Note your Project name and connected Foundry resource name
2	Model catalog	Search "gpt-4o" — note the publisher shown
3	Deployments tab	Confirm your GPT-4o deployment is listed and shows "Succeeded"
4	Chat Playground	Locate the System message box and the model/deployment selector
5	Management Center → Quota	Note remaining TPM quota for GPT-4o in your region
6	Management Center → Connected resources	Identify the Azure OpenAI / AI Services resource behind your project
7	Access control (IAM)	Confirm your identity shows Cognitive Services OpenAI User on this resource

Exercise 3 — Model Selection Decision Exercise

Goal: apply a repeatable decision framework instead of defaulting to "always pick the biggest model." For each BFL group-company scenario below, decide which model characteristic matters most, and record your recommended model choice and deployment type.

Decision pointers to weigh for every scenario: cost per token, context window needed, multimodal (vision) requirement, latency/throughput sensitivity, and reasoning complexity.

Scenario	Business Need	Key Decision Pointers	Your Pick
A — AMC (Asset Mgmt.)	Analysts need monthly fund fact-sheets summarized; occasionally must read tables/charts from scanned PDF reports.	Needs vision, accuracy matters more than raw speed, moderate volume	
B — BHFL (Housing Finance)	Public website FAQ chatbot answering routine eligibility/tenure questions at high concurrency.	High volume, cost-sensitive, simple Q&A, no images	
C — BLIL (Life Insurance)	Claims desk assistant reads scanned claim forms and photos of supporting documents to extract key details.	Needs strong vision + reasoning accuracy, lower volume	
D — BFSL (Securities)	Trade-desk assistant must answer short, structured compliance-rule lookups in near real time.	Latency-critical, simple lookups, cost-sensitive	

Be ready to justify your answer for at least one scenario when your trainer opens discussion — there is more than one defensible answer depending on how you weigh cost against accuracy.

Exercise 4 — Chat Playground: Scenario-Based Prompting

Goal: experience firsthand why a capable model still fails without the right information.

- Open the Chat Playground in your project and select your `trn_<##>-gpt4o` deployment.
- Without adding any system instructions or reference material, ask: "What is the maximum loan tenure for the BFL SecureHome Loan?"
- Record the model's response. Notice whether it guesses, hallucinates a plausible-sounding number, or admits it doesn't know.
- Ask a second question: "What is the prepayment charge on the BFL FlexiBiz Working Capital Loan if foreclosed within 12 months?" Record this response as well.

Exercise 5 — System Instructions Tuning

Goal: see what system instructions can and cannot fix on their own.

- In the System message box, add: "You are the BFL Loan Product Assistant. Only answer using information explicitly provided to you in this conversation. If the information has not been provided, say 'I don't have that information in my knowledge base' rather than guessing."
- Re-ask the same two questions from Exercise 4.
- Compare the new responses to your Exercise 4 notes. The model should now decline rather than guess — but it still can't answer correctly, because instructions alone don't supply missing facts.

Exercise 6 — Manual RAG with Synthetic Documents

Goal: manually ground the assistant, and personally feel the scaling limitation that Session 6 solves with automated RAG.

- Open `BFL_SecureHome_Loan_Policy.txt` and copy its full contents.

- Paste the contents into the chat as context (either appended to the system message or as a user message stating "Use the following reference document to answer questions:"), then re-ask the SecureHome tenure question from Exercise 4.
- Confirm the assistant now answers correctly, citing the tenure figure from the document.
- Without pasting the FlexiBiz document, re-ask the FlexiBiz prepayment-charge question. Confirm the assistant correctly declines — because only one document was ever placed in its context window.

The scaling problem

You just proved grounding works — and that pasting documents by hand doesn't scale past one or two files, let alone an entire product catalog. Session 6 replaces this manual copy-paste step with Azure AI Search and automated retrieval, so the right document is found and injected automatically, for any question, across the full document set.

Wrap-Up

You've now provisioned a Foundry project end-to-end, made a structured model choice instead of a default one, and personally experienced the difference between an ungrounded, instructed, and manually grounded assistant. That progression — ungrounded → instructed → grounded — is the mental model to carry into every AI architecture conversation from here forward, including Session 6's deep dive into Foundry, models, RAG, and agents.

References

- Microsoft Learn — Create a Foundry project: <https://learn.microsoft.com/azure/foundry/how-to/create-projects>
- Microsoft Learn — Foundry Models sold by Azure (region availability): <https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/models-sold-directly-by-azure-region-availability>
- Microsoft Learn — Role-based access control for Azure OpenAI: <https://learn.microsoft.com/en-us/azure/ai-foundry/openai/how-to/role-based-access-control>
- Microsoft Learn — Azure OpenAI quotas and limits: <https://learn.microsoft.com/azure/ai-foundry/openai/quotas-limits>